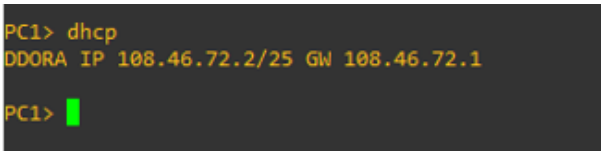
ΔΙΚΤΥΑ ΥΠΟΛΟΓΙΣΤΩΝ (6η Εργαστηριακή Άσκηση)

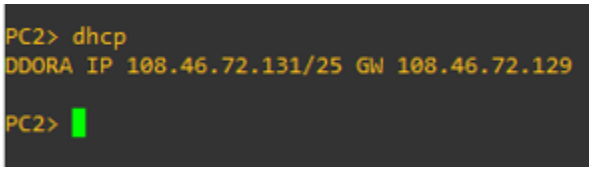
ΕΠΩΝΗΜΟ:ΚΙΟΥΡΤ ΜΕΧΜΕΤΑΛΗ

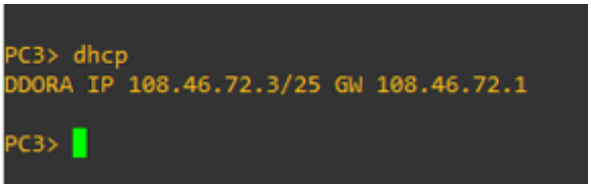
ΟΝΟΜΑ:ΑΧΜΕΤ

ΑΜ:1084672

Αρχικά εκτελώντας DHCP βλέπουμε ότι έχουμε επικοινωνία με το έξω κόσμο.

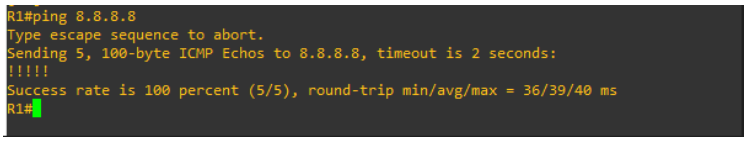


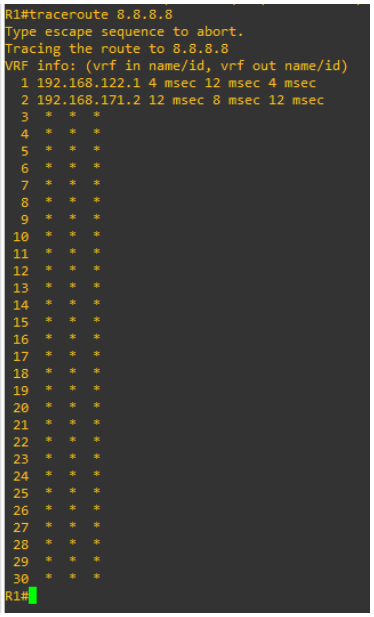




Ερωτήσεις

1η Ερώτηση: Παρατηρούμε ότι κατεβάζει data με αποτέλεσμα να υπάρχει πρόσβαση με το εξωτερικό δίκτυο.





2η Ερώτηση :

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
108.46.72.2         0100.5079.6668.00       May 27 2023 12:33 PM    Automatic
108.46.72.3         0100.5079.6668.02       May 27 2023 12:34 PM    Automatic
108.46.72.130       ff72.9c00.0000.0100.    May 26 2023 12:45 PM    Automatic
                    012c.027d.ab0c.7872.
                    9c00.00
108.46.72.131       0100.5079.6668.01       May 27 2023 12:33 PM    Automatic
R1#
```

3η Ερώτηση:

Παρατηρούμε ότι έχει πάρει τα ονόματα που είχαμε ορίσει με αποτέλεσμα να μας βγάζει το πίνακα με τις ρυθμίσεις.

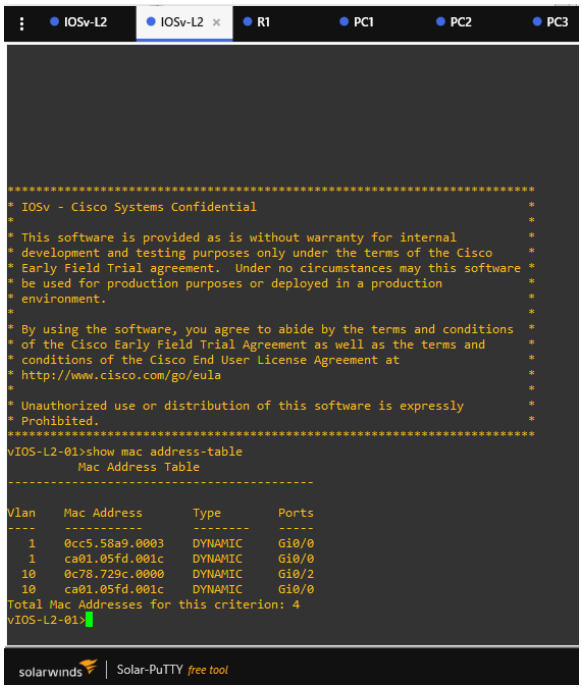
```
R1#show ip dhcp pool

Pool 1084672-1DHCP :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 2
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 108.46.72.4          108.46.72.1      - 108.46.72.126     2

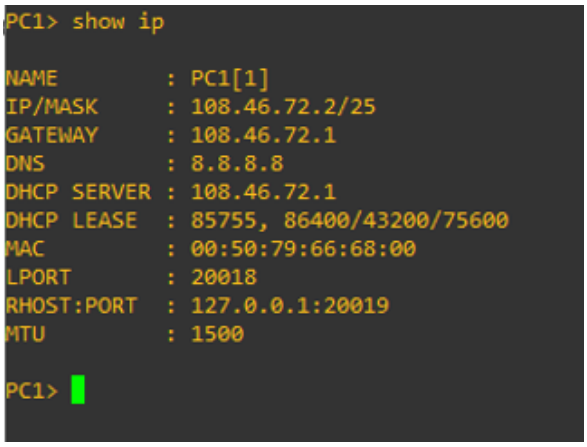
Pool 1084672-2DHCP :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 2
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 108.46.72.132        108.46.72.129    - 108.46.72.254     2
R1#
```

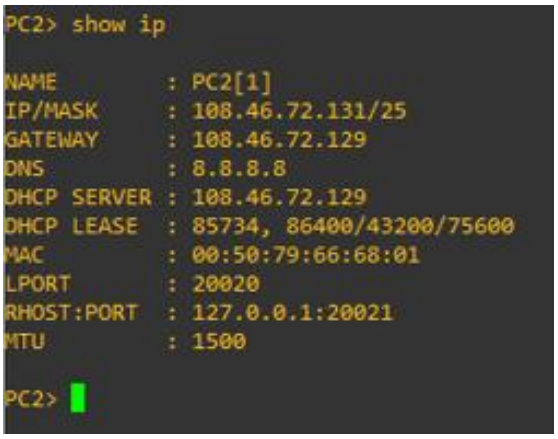
4η Ερώτηση:

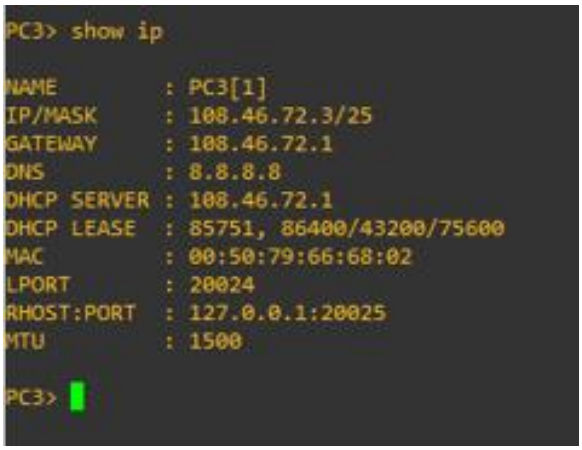




5η Ερώτηση: Έχουμε τα αποτελέσματα με τα IP αντίστοιχα με κάθε PC







```
debian@debian:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 0c:78:72:9c:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s4
    inet6 fe80::e78:72ff:fe9c:0/64 scope link
       valid_lft forever preferred_lft forever
debian@debian:~$
```

6η Ερώτηση:

Για το PC1:

```
PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=127 time=59.382 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=127 time=57.343 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=127 time=56.344 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=127 time=51.531 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=127 time=54.114 ms
```

```
PC1> ping google.com
google.com resolved to 142.251.209.14

84 bytes from 142.251.209.14 icmp_seq=1 ttl=127 time=62.326 ms
84 bytes from 142.251.209.14 icmp_seq=2 ttl=127 time=58.253 ms
84 bytes from 142.251.209.14 icmp_seq=3 ttl=127 time=55.294 ms
84 bytes from 142.251.209.14 icmp_seq=4 ttl=127 time=50.864 ms
84 bytes from 142.251.209.14 icmp_seq=5 ttl=127 time=62.426 ms

PC1>
```

Για το PC2:

```
PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=127 time=71.029 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=127 time=54.843 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=127 time=50.299 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=127 time=48.441 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=127 time=47.393 ms

PC2> ping google.com
google.com resolved to 142.251.209.14

84 bytes from 142.251.209.14 icmp_seq=1 ttl=127 time=40.234 ms
84 bytes from 142.251.209.14 icmp_seq=2 ttl=127 time=78.173 ms
84 bytes from 142.251.209.14 icmp_seq=3 ttl=127 time=51.529 ms
84 bytes from 142.251.209.14 icmp_seq=4 ttl=127 time=56.960 ms
84 bytes from 142.251.209.14 icmp_seq=5 ttl=127 time=69.386 ms

PC2>
```

Για το PC3:

```
PC3> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=127 time=65.835 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=127 time=64.609 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=127 time=60.189 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=127 time=62.278 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=127 time=104.361 ms

PC3> ping google.com
google.com resolved to 142.251.209.14

84 bytes from 142.251.209.14 icmp_seq=1 ttl=127 time=61.358 ms
84 bytes from 142.251.209.14 icmp_seq=2 ttl=127 time=64.272 ms
84 bytes from 142.251.209.14 icmp_seq=3 ttl=127 time=68.795 ms
84 bytes from 142.251.209.14 icmp_seq=4 ttl=127 time=83.352 ms
84 bytes from 142.251.209.14 icmp_seq=5 ttl=127 time=82.894 ms

PC3>
```

Για το Debian:

```
traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  static-108-46-72-129.nycmny.fios.verizon.net (108.46.72.129)  28.819 ms  26.429 ms  25.914 ms
 2  192.168.122.1 (192.168.122.1)  34.603 ms  34.244 ms  33.754 ms
 3  192.168.171.2 (192.168.171.2)  33.347 ms  32.808 ms  32.362 ms
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
debian@debian:~$
```

```
debian@debian:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=59.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=50.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=49.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=127 time=69.4 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=127 time=53.4 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=127 time=53.3 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=127 time=63.3 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=127 time=52.2 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=127 time=49.7 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=127 time=63.9 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=127 time=64.8 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=127 time=48.9 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=127 time=52.0 ms
^C
--- 8.8.8.8 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12018ms
rtt min/avg/max/mdev = 48.884/56.265/69.394/6.708 ms
debian@debian:~$
```

```
debian@debian:~$ ping google.com
PING google.com (142.251.209.14) 56(84) bytes of data.
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=1 ttl=127 time=69.7 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=2 ttl=127 time=46.3 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=3 ttl=127 time=53.7 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=4 ttl=127 time=41.4 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=5 ttl=127 time=46.9 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=6 ttl=127 time=48.9 ms
64 bytes from mil04s50-in-f14.1e100.net (142.251.209.14): icmp_seq=7 ttl=127 time=48.9 ms
```

7η Ερώτηση: Βλέπουμε ότι για το PC1 και για το PC3 χρειάζεται 2 βήματα για να φτάσει στο δικό του VLAN και σε διαφορετικό VLAN χρειάζεται 1 βήμα. Στο PC2 χρειάζεται μόνο 1 βήμα.

```
debian@debian:~$ traceroute 108.46.72.2
traceroute to 108.46.72.2 (108.46.72.2), 30 hops max, 60 byte packets
 1  static-108-46-72-129.nycmny.fios.verizon.net (108.46.72.129)  24.082 ms  25.352 ms  23.979 ms
 2  static-108-46-72-2.nycmny.fios.verizon.net (108.46.72.2)  62.199 ms  61.183 ms  60.447 ms
debian@debian:~$ traceroute 108.46.72.131
traceroute to 108.46.72.131 (108.46.72.131), 30 hops max, 60 byte packets
 1  static-108-46-72-131.nycmny.fios.verizon.net (108.46.72.131)  17.057 ms  15.663 ms  15.657 ms
debian@debian:~$ traceroute 108.46.72.3
traceroute to 108.46.72.3 (108.46.72.3), 30 hops max, 60 byte packets
 1  static-108-46-72-129.nycmny.fios.verizon.net (108.46.72.129)  19.979 ms  28.565 ms  28.462 ms
 2  yonkers.bigcityautomotive.com (108.46.72.3)  33.345 ms  33.271 ms  33.206 ms
```

Υποβολή Εργασίας

Για κάποιο λόγο αυτό μου βγάζει για να κάνω την υποβολή και δυστυχώς δεν μπορεί να γίνει το link.

```
debian@debian:~$ sudo apt-get install update
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package update
debian@debian:~$ sudo apt-get install telnet gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gcc is already the newest version (4:10.2.1-1).
telnet is already the newest version (0.17-42).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
debian@debian:~$
```

Τελικά γίνεται και η υποβολή ως SUCCESS

